

Priority Queue – 5 Worked Examples

A **Priority Queue** is a queue in which each element is associated with a **priority**. The element with the **highest priority is served first**, regardless of when it was inserted.

In the examples below, **smaller priority number = higher priority**:

Priority 1 > Priority 2 > Priority 3 > Priority 4

Example 1: Basic Priority Queue Insertion

Suppose we have:

Element	Priority
A	3
B	1
C	2
D	4

Insert in this order:

A → B → C → D

After arranging according to priority:

Position	Element	Priority
0	B	1
1	C	2
2	A	3
3	D	4

So the **deletion order** will be:

B → C → A → D

FRONT and REAR

FRONT



[B, 1] → [C, 2] → [A, 3] → [D, 4]

REAR



Therefore:

FRONT = 0

REAR = 3

Example 2: Enqueue and Dequeue

Consider a priority queue of size 5.

Operations:

ENQUEUE (10, 3)

ENQUEUE (20, 1)

ENQUEUE (30, 2)

DEQUEUE ()

ENQUEUE (40, 1)

DEQUEUE ()

Here the second value is the priority.

Step-by-step

Step	Operation	Queue	FRONT	REAR
0	Initial	[]	-1	-1
1	Enq(10,3)	[(10,3)]	0	0
2	Enq(20,1)	[(20,1), (10,3)]	0	1
3	Enq(30,2)	[(20,1), (30,2), (10,3)]	0	2
4	Dequeue	[(30,2), (10,3)]	0	1
5	Enq(40,1)	[(40,1), (30,2), (10,3)]	0	2
6	Dequeue	[(30,2), (10,3)]	0	1

Notice

First:

20 has priority 1

so 20 is removed.

Then 40 is inserted with priority 1, so it becomes the next element to be removed.

Final deletion order:

20 → 40 → 30 → 10

Example 3: Same Priority

Suppose **larger number = higher priority** in this example.

Element	Priority
A	5
B	2
C	5
D	3

If two elements have the same priority, we use **FIFO order** between them.

So:

A(5) → C(5) → D(3) → B(2)

Queue

FRONT

↓

[A, 5] → [C, 5] → [D, 3] → [B, 2]

↑

REAR

Therefore:

```
FRONT = 0
REAR  = 3
```

The important point is:

```
A and C have same priority
```

Since A was inserted first, **A is removed before C**.

This is called **FIFO among equal-priority elements**.

Example 4: Hospital Emergency Priority Queue

Consider a hospital emergency department.

Priority:

```
1 = Critical
2 = Serious
3 = Normal
4 = Minor
```

Patients arrive as follows:

Patient	Priority
P1	3
P2	1
P3	4
P4	2
P5	1

After arranging by priority:

```
P2 (1) → P5 (1) → P4 (2) → P1 (3) → P3 (4)
```

Queue representation

```
FRONT
↓
```

P2, 1	P5, 1	P4, 2	P1, 3	P3, 4
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↑
REAR

Therefore:

```
FRONT = 0
REAR  = 4
```

Deletion sequence

If doctors attend patients according to priority:

```
DEQUEUE() → P2
DEQUEUE() → P5
DEQUEUE() → P4
DEQUEUE() → P1
DEQUEUE() → P3
```

Notice that **P5**, although arriving later, is served before P4 because P5 has priority **1**.

Example 5: Complete Priority Queue Computation

Assume:

```
MAX = 6
Smaller priority number = higher priority
```

Perform:

```
ENQUEUE(10, 3)
ENQUEUE(20, 1)
ENQUEUE(30, 4)
ENQUEUE(40, 2)
DEQUEUE()
ENQUEUE(50, 1)
DEQUEUE()
ENQUEUE(60, 3)
DEQUEUE()
```

Step-by-step table

Step	Operation	Queue after operation	FRONT	REAR
0	Initial	[]	-1	-1
1	Enq(10,3)	10 (3)	0	0
2	Enq(20,1)	20 (1) , 10 (3)	0	1
3	Enq(30,4)	20 (1) , 10 (3) , 30 (4)	0	2
4	Enq(40,2)	20 (1) , 40 (2) , 10 (3) , 30 (4)	0	3
5	Dequeue	40 (2) , 10 (3) , 30 (4)	0	2
6	Enq(50,1)	50 (1) , 40 (2) , 10 (3) , 30 (4)	0	3
7	Dequeue	40 (2) , 10 (3) , 30 (4)	0	2
8	Enq(60,3)	40 (2) , 10 (3) , 60 (3) , 30 (4)	0	3
9	Dequeue	10 (3) , 60 (3) , 30 (4)	0	2

Final queue

FRONT

↓

10, 3	60, 3	30, 4
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↑

REAR

So:

FRONT = 0

REAR = 2

The next element to be deleted is:

10

because 10 and 60 both have priority 3, and 10 arrived first.

Priority Queue – Important Rules

1. Insertion

A new element is inserted according to its priority.

Example:

Current:

20 (1) → 30 (2) → 40 (4)

Insert:

10 (3)

Result:

20 (1) → 30 (2) → 10 (3) → 40 (4)

2. Deletion

Always remove the element with the **highest priority**.

20 (1) → 30 (2) → 10 (3)

↑

DELETE

20 is deleted.

3. Equal Priority

If two elements have the same priority, normally use **FIFO**.

A (2) → B (2) → C (3)

A is deleted before B.

Priority Queue vs Simple Queue

Simple Queue

FIFO

First inserted → first deleted

A B C D

One ordering criterion

Example: ticket counter

Priority Queue

Based on priority

Highest priority → first deleted

B A D C depending on priority

Priority + FIFO for ties

Example: emergency room

Easy exam example

If students are given:

A(3), B(1), C(4), D(2), E(1)

with **1 = highest priority**, ask them to determine:

Insertion order:

A → B → C → D → E

Priority order:

B(1) → E(1) → D(2) → A(3) → C(4)

Deletion order:

B → E → D → A → C

This is a good example for students to practice **FRONT, REAR, priority comparison, insertion, deletion, and FIFO handling of equal priorities.**

Is this conversation helpful so far?